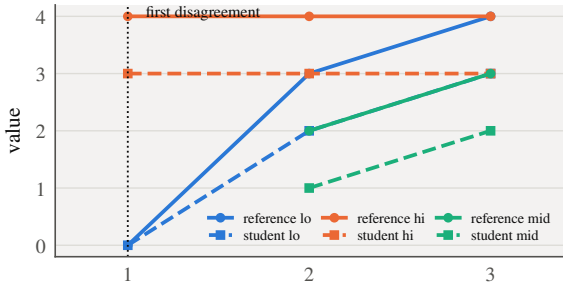


lower_bound([1, 3, 5, 7], 8): the student's hi starts at $\text{len}(a) - 1$



loop-invariant checkpoint (each evaluation of the loop guard)